

Exploring the Space of 2-by-2 Games

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Fishing Game

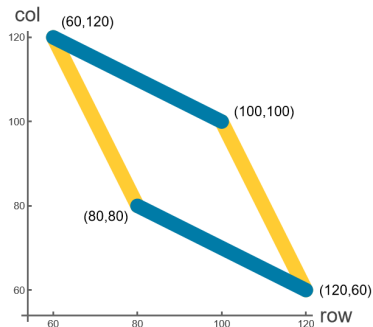
- Two fishermen fish a lake.
- They each may catch either 10 fish or 20 fish per year.
- If both fishermen catch 10 fish, the fish will be gone after 10 years.
- If one fisherman catches 20 fish, the fish will be gone after 6 years.
- If both fishermen catch 20 fish, the fish will be gone after 4 years.
- How should the fishermen fish?

$$R = \begin{array}{cc} & \begin{array}{c} 10 \\ 20 \end{array} \\ \begin{array}{c} 10 \\ 20 \end{array} & \begin{pmatrix} 100 & 60 \\ 120 & 80 \end{pmatrix} \end{array} \quad \begin{array}{c} 10 \\ 20 \end{array} \quad C = \begin{array}{cc} & \begin{array}{c} 10 \\ 20 \end{array} \\ \begin{array}{c} 10 \\ 20 \end{array} & \begin{pmatrix} 100 & 120 \\ 60 & 80 \end{pmatrix} \end{array} \quad \begin{array}{c} 10 \\ 20 \end{array}$$

Diagrams

$$R = \begin{pmatrix} 100 & 60 \\ 120 & 80 \end{pmatrix} \quad C = \begin{pmatrix} 100 & 120 \\ 60 & 80 \end{pmatrix}$$

- We plot the points (100, 100), (60, 120), (120, 60), and (80, 80).
- Points from the same row are connected by a cerulean line.
- Points from the same column are connected by a gold line.

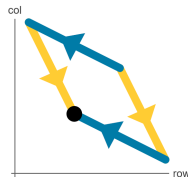
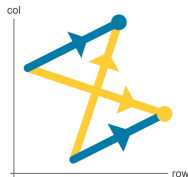
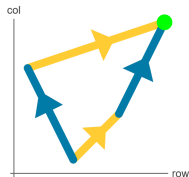
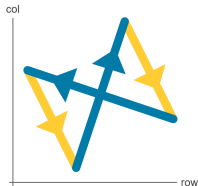


Equilibria

Definition

An *equilibrium* exists when neither player can get a better payoff by changing their choice.

We add a green, gold, cerulean, or black dot to a game's diagram where it has an equilibrium.



A Four-Dimensional Space

- At its most general, a game has eight degrees of freedom.
- Adding a constant to either matrix makes no impact.
- Multiplying either matrix by a positive constant makes no impact.
- We normalize matrices to range from 0 to 6.

$$R_1 = \begin{pmatrix} 100 & 60 \\ 120 & 80 \end{pmatrix}$$

$$C_1 = \begin{pmatrix} 100 & 120 \\ 60 & 80 \end{pmatrix}$$

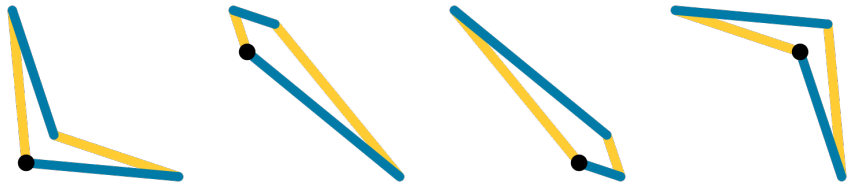
$$R_2 = \begin{pmatrix} 4 & 0 \\ 6 & 2 \end{pmatrix}$$

$$C_2 = \begin{pmatrix} 4 & 6 \\ 0 & 2 \end{pmatrix}$$

- The resulting four-dimensional space can be divided into 576 cells.

Categorizing Games

- Each matrix can be ordered in $4! = 24$ ways.
- That's $24^2 = 576$ types of game.
- However, swapping the rows or columns of both matrices results in a functionally identical game.
- This brings us to $24^2/2^2 = 144$ cells.
- There is one more relevant consideration...

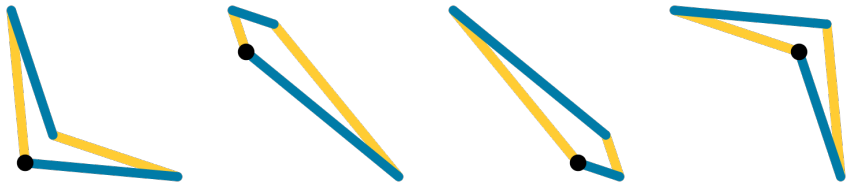


Temperature

Definition

The *temperature* of a matrix is the mean of its second and third-greatest entries.

- Matrices with a low temperature are called *cool*.
- Matrices with a high temperature are called *warm*.
- A game can be cool-cool, cool-warm, warm-cool, or warm-warm.
- By breaking each cell into four smaller cells according to temperature, we get $144 \times 4 = 576$ cells.



Bibliography

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- Bryan Bruns. *Atlas of 2×2 Games: Transforming Conflict and Cooperation*, <https://dlc.dlib.indiana.edu/dlc/items/224ba16c-524d-4f7a-a18d-9f429b082aee>, (2015). Retrieved October 31, 2025.